ELEC3104: Digital Signal Processing

Topic 3: Digital Filter Fundamentals

Tutorial questions

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Topic 3: Pass-Level Tutorial Questions

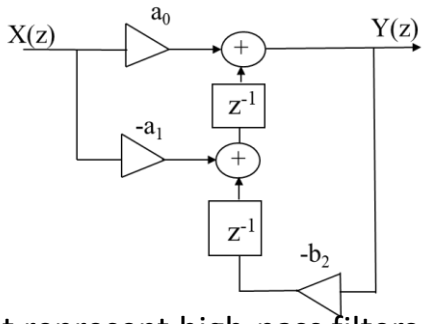
1. A digital filter structure is shown below. Calculate the dc gain of the filter, in terms of the filter coefficients. (Tick the correct answer)

   (a) $1 + a_0 - a_1$

   (b) $\frac{a_0 - a_1}{1 + b_2}$

   (c) $\frac{a_0 - a_1}{1 - b_2}$

   (d) Non of the above



2. Tick all transfer functions that represent high-pass filters, where $0 < \alpha < 1$.

   $(a)\ \ H(z) = 1 - \alpha\, z^{-1}$ $\quad (b)\ \ H(z) = \frac{1}{1 - \alpha\, z^{-1}}$

   $(c)\ \ H(z) = 1 + \alpha\, z^{-1}$ $\quad (d)\ \ H(z) = \frac{1}{1 + \alpha\, z^{-1}}$

3. The Digital Filter $H(z) = \frac{2 + 2z^{-1}}{3 - z^{-1}}$; (Tick the correct answer)

   a) is a low-pass filter
   b) is a high-pass filter
   c) is an all-pass filter

4. A maximum- phase FIR filter has: (Tick the correct answer)

   a) All zeros and poles outside the unit circles
   b) All poles outside the unit circle
   c) All zeros outside the unit circle
   d) All poles and zeros inside the unit circle

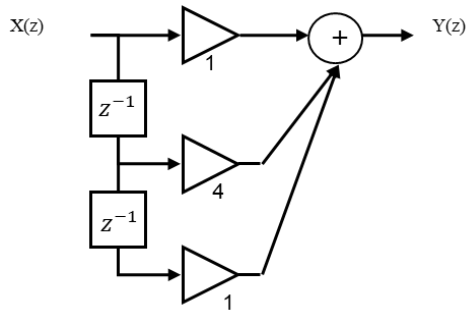
5. Which of the following are linear- phase filters: (Tick the correct answer)

   (a) $H(z) = 0.25 + 0.5\, z^{-1} + 0.25\, z^{-2}$ (b) $H(z) = z^{-4}$

   (c) $H(z) = 1 - z^{-2}$ (d) $H(z) = 1 + 2z^{-1} + 2\, z^{-2} - z^{-3}$

6. What can be said about the following filter ? (Tick the correct answer)

   a) It has minimum phase and a finite impulse response.
   b) It has linear phase and an infinite impulse response.
   c) It has an infinite impulse response.
   d) It has linear phase and a finite impulse response



7. Which of the following system is linear, time - invariant, causal and stable? (Tick the correct answer)

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1. Compute the 3-dB bandwidth of the following filters:

$$H_1(z) = \frac{1-a}{1-az^{-1}}; \quad H_2(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}, \qquad 0 < a < 1$$

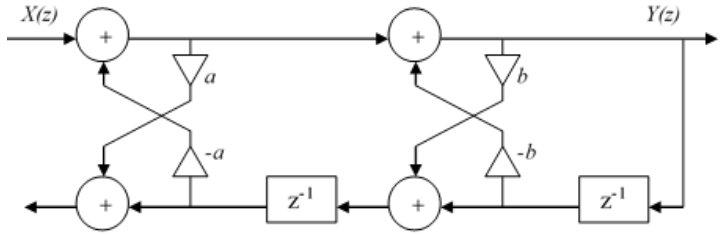
Which filter has a smaller 3dB bandwidth?

2. Find the magnitude response of the filter *H(z)* and comment on your result.

$$H(z) = \frac{1+3z^2}{3+z^2}$$

Draw the canonic realisation of the filter *H(z)*.

3. Determine the frequency response $H(\theta)$ of the filter represented by the following structure:



4. A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1-b_1z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad |H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$$

5. Implement for the following *H(z) using* a parallel structure"

$$H(z) = \frac{z^2-4z+3}{z(z+0.5)^2}$$

6. A notch filter is given by $H(z) = \frac{1+z^{-2}}{1+0.8z^{-2}}$. Determine the magnitude response at $\theta = 0, \frac{\pi}{2}, and\ \pi$. Sketch the magnitude response in the interval $-\pi \le \theta \le \pi$.

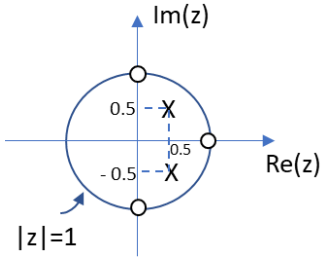
7. Determine and sketch the magnitude and phase responses of the following filters

(a) $y[n] = \frac{1}{2}(x[n] - x[n-1])$ (b) $y[n] = x[n] - x[n-8]$

(c) $y[n] = x[n-4]$

8 (a) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown below:

(b) Using the pole-zero map , sketch the approximate magnitude response $|H(\theta)|$.



9. A finite impulse filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$. Determine b_4 to b_6 such that the filter has linear phase.

10. Implement the following transfer function using a cascade structure, direct form II and in parallel structure. Assume all coefficients are real and $a \neq -b$.

$$H(z) = \frac{1}{(1+az^{-1})(1-bz^{-1})}$$

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11. Show that both digital filters given below have the same magnitude response:

(a) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-i]$ (b) $y[n] = \sum_{i=-m}^{m} c_i\, x[n-m-i]$

where $y[n]$ is the output, $x[n]$ is the input and c_i are the coefficients.

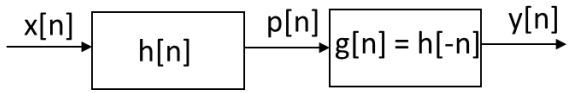
12. Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2+z+0.5)}$$

Is *H(z)* a stable function? If it is not stable, find a stable transfer function *G(z)* such that $|G(\theta)| = |H(\theta)|$.

13. Show that for a real valued impulse response *h*[n] the structure below has a zero − phase frequency Response.

$$H(z) = (\tfrac{1}{2} + z^{-1} + \tfrac{1}{2}z^{-2}).$$



14. The difference equation of a digital filter is given by

$$y[n] = (1/8)\{x[n] - x(n-8)\} - y(n-2)$$

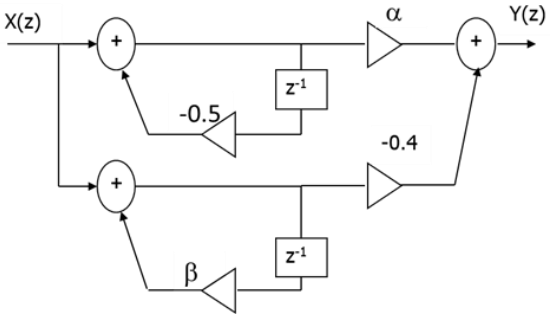
(a) Find the transfer function for the above filter.

(b) Using the transfer function obtained in part (a) above, draw the pole-zero map for the filter and sketch the corresponding magnitude response.

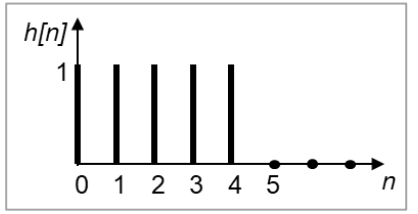
15. The following figure shows an IIR filter structure, and its transfer function given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3z(5z-2)}{(z+0.5)(3z-1)}$$

Determine the values of the multiplier coefficients α and β.



16. A FIR filter has an impulse response as shown below:

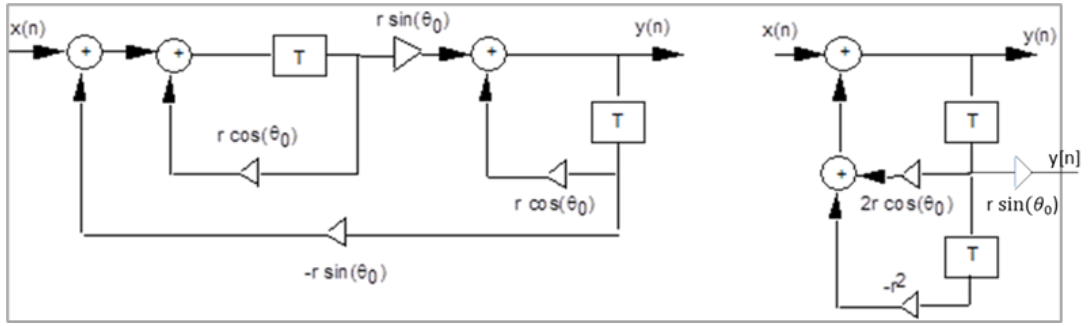


(a) Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$

(b) Draw an implementation of this filter as a cascade structure.

17. Show that the systems (i) and (ii) shown below are equivalent:



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18. A third order FIR filter has the transfer function G(z) given by

$$G(z) = (9 + 6z^{-1} - 8z^{-2})(2 + 5z^{-1}).$$

Determine FIR transfer functions having the same magnitude response as G(z) that are:

(a) minimum phase
(b) maximum phase

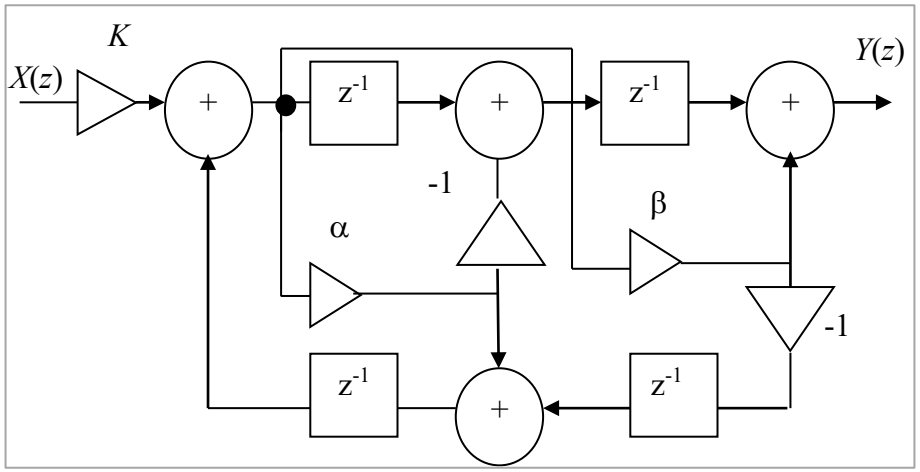
19. Analyse the digital filter structure shown on the right:

(a) Determine its transfer function $H(z)$ =Y(z)/X(z)

(b) Determine the value of K such that $H(z)$ has unity gain at $\theta = 0$.

(c) Determine the value of K such that $H(z)$ has unity gain at $\theta = \pi$.

(d) Are the values of K obtained in (b) and (c) different? Explain your answer.



20. (a) Calculate the magnitude and phase responses of the following FIR filter:

$$y[n] = \frac{1}{4}[x[n] - 2x[n-1] + x[n-2]]$$

(b) Sketch the magnitude and phase responses.
(c) Show that the filter has a constant group delay and determine its value.

Topic 3: Higher–Level Tutorial questions

1. The frequency response of an ideal differentiator is given by

$$H(\theta) = j\theta e^{-j\theta\tau} \text{ - } \pi \leq \theta \leq \pi$$

Note: $\int \theta\, e^{jk\theta} d\theta = \frac{e^{jk\theta}}{k^2}(1 - jk\theta)$

This response is periodic with period 2π. The quantity τ is the delay of the system in samples.

   (a) Sketch the magnitude and phase responses of the system for $-\pi \leq \theta \leq \pi$.

   (b) Find Find the impulse response $h[n]$ of the system, expressing your answer in terms of sine and cosine functions.

2. Determine the magnitude and phase responses of the multipath channel *y[n] = x[n] + x[n-M]*. At what frequencies $|H(\theta)| = 0$?

3. Let a causal linear time invariant discrete-time system be characterised by a real impulse response *h*[*n*] with a DTFT of $H(\theta)$. Consider the system shown in Figure A , where *x*[*n*] is a finite length sequence. Determine the frequency response of the overall system $G(\theta)$ $\{G(z) = {}^{Y(z)}/_{X(z)}\}$ in terms of $H(\theta)$ and show that it has a zero-phase response.

4. An analogue system may be implemented as shown in Figure B. Using the transformation $\frac{1}{s} \rightarrow k\frac{1+z^{-1}}{1-z^{-1}}$ a discrete time system may be obtained; where k is a constant.

   (a) Draw the digital structure of the system, using the minimum number of multipliers and delay units .

   (b) Derive the difference equation governing the system in the form $y[n] = 2\cos[\theta_0]\, y[n-1] - y[n-2]$ and determine cos (θ_0).

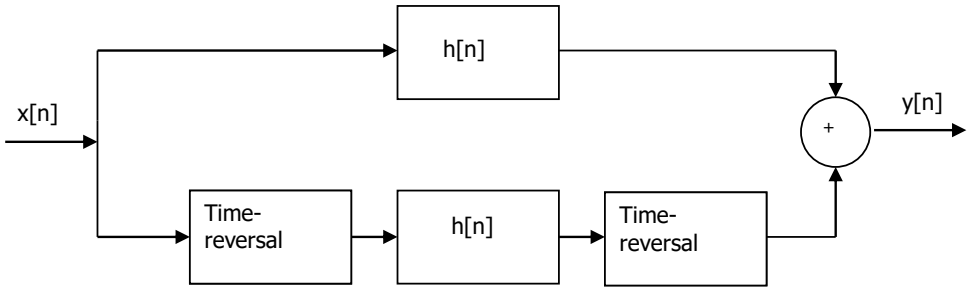


Figure A

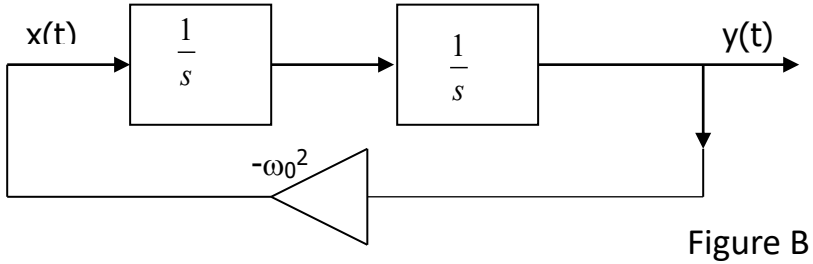


Figure B

Topic 3: Higher–Level Tutorial questions

5. (a) Find the group delay for each of the following systems, where $|\alpha| < 1$

(i) $H_1(\theta) = 1 - \alpha e^{-j\theta}$ (ii) $H_2(\theta) = \frac{1}{1-\alpha e^{-j\theta}}$

(b) Using the results obtained in (a), show that the group delay of the all-pass filter with a transfer function $H_0(z) = \frac{z^{-1}-\alpha}{1-\alpha z^{-1}}$ is $\frac{1-\alpha^2}{1+\alpha^2-2\alpha\cos(\theta)}$

6. Find the minimum-phase system having the following magnitude - squared response:

$$|H(\theta)|^2 = \frac{\frac{5}{4}-\cos\theta}{\frac{10}{9}-\frac{2}{3}\cos\theta}$$

7. A stable and causal system can be decomposed into a minimum-phase system and an all-pass system: $H(z) = H_{min}(z) \cdot H_{ap}(z)$

Consider the following system : $H(z) = \frac{(1-0.7z^{-1})(1-j2z^{-1})(1+j2z^{-1})}{(1-0.8z^{-1})(1+0.8z^{-1})}$

(a) Find $H_{min}(z)$ and $H_{ap}(z)$,

(b) Determine whether H(z) can be expressed as $H(z) = H_{min}(z) \cdot H_{lp}(z)$ where $H_{lp}(z)$ is a linear phase system and if so, find the two components.

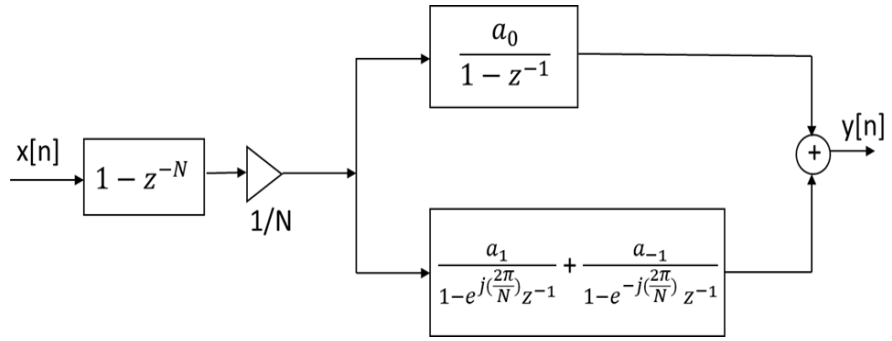
8 A filter structure with a transfer function $\{H(z) = \frac{Y(z)}{X(z)}\}$ is shown below. You may assume that $N = 3$, $a_0 = 3$ and $H(\theta)|_{\theta=2\pi/3} = 6+j3\sqrt{3}$.

Also $H\left(e^{j\frac{2\pi k}{N}}\right) = a_k$ and $H(e^{-j\frac{2\pi k}{N}}) = a_{-k}$

(a) Determine a_1 & a_{-1} such that this filter has a real - valued impulse response *h[n];*

(b) Find H(z) and draw a filter structure using delay elements, multipliers and adders.

(c) Show that the filter can be realised as an FIR filter.



(d)